

SUBJECT 1: Information & Cyber Security

◆ Unit I: Introduction to Security (20 Questions + Answers)

1. What is Information Security?

Information security is the practice of protecting information from unauthorized access, use, disclosure, disruption, modification, or destruction. It ensures data confidentiality, integrity, and availability (CIA triad).

2. Explain the CIA Triad.

- **Confidentiality:** Only authorized users can access data.
 - **Integrity:** Data remains accurate and unchanged.
 - **Availability:** Data is accessible when needed.
-

3. What is a Security Policy?

A security policy is a formal document defining rules and procedures to protect an organization's data and systems. It specifies user roles, access control, and security measures.

4. What are Threats?

A threat is any potential danger that can harm a system or data, such as hackers, malware, or natural disasters.

5. What are Vulnerabilities?

A vulnerability is a weakness in a system that can be exploited by attackers, such as weak passwords or outdated software.

6. What is an Attack?

An attack is an attempt to exploit a vulnerability to gain unauthorized access or damage a system.

7. Types of Security Attacks

- **Passive:** Monitoring data (eavesdropping)
 - **Active:** Modifying or disrupting data
-

8. What is Network Security?

Network security involves protecting a network from unauthorized access, misuse, or attacks.

9. What are Security Services?

Security services include authentication, authorization, confidentiality, integrity, and non-repudiation.

10. What is Authentication?

Authentication verifies the identity of a user using passwords, biometrics, or tokens.

11. What is Authorization?

Authorization determines what resources an authenticated user can access.

12. What is Non-repudiation?

It ensures that a sender cannot deny sending a message.

13. What is Confidentiality?

Confidentiality ensures that information is accessible only to authorized users.

14. What is Integrity?

Integrity ensures that data is not altered or tampered with.

15. What is Availability?

Availability ensures systems and data are accessible when required.

16. What is Risk in Security?

Risk is the potential loss due to a threat exploiting a vulnerability.

17. What is Risk Management?

Process of identifying, analyzing, and mitigating risks.

18. What is Cyber Attack?

An attempt to damage or gain unauthorized access to computer systems.

19. What is Malware?

Malicious software like viruses, worms, and trojans designed to harm systems.

20. What is Phishing?

A technique where attackers trick users into revealing sensitive information via fake emails or websites.

◆ Unit II: Data Encryption Techniques (20 Questions)

1. What is Cryptography?

Technique used to secure communication by converting plaintext into ciphertext.

2. What is Encryption?

Process of converting readable data into unreadable form.

3. What is Decryption?

Process of converting ciphertext back to plaintext.

4. What is Symmetric Encryption?

Uses a single key for encryption and decryption.

5. What is Asymmetric Encryption?

Uses a pair of keys: public and private.

6. Difference between Symmetric & Asymmetric

Symmetric is faster; asymmetric is more secure for key exchange.

7. What is DES?

A symmetric encryption algorithm with 56-bit key, now considered insecure.

8. What is AES?

Modern encryption standard with strong security and key sizes of 128, 192, and 256 bits.

9. What is Triple DES?

Applies DES three times for improved security.

10. What is Block Cipher?

Encrypts data in fixed-size blocks.

11. What is Stream Cipher?

Encrypts data one bit/byte at a time.

12. What is Cryptanalysis?

Study of breaking encryption.

13. What is Key?

A value used in encryption/decryption process.

14. What is Plaintext?

Original readable data.

15. What is Ciphertext?

Encrypted unreadable data.

16. What is Substitution Cipher?

Replaces characters with other characters.

17. What is Transposition Cipher?

Rearranges characters without changing them.

18. What is Brute Force Attack?

Trying all possible keys to decrypt data.

19. What is Weak Key?

A key that produces predictable encryption results.

20. What are Limitations of DES?

Small key size, vulnerable to brute force attacks.

👉 This is already a lot to revise properly.

👉 **Next Step**

I'll continue with:

- Unit III, IV, V, VI (Security)
- Then full Cloud Computing units

Just say “continue” and I'll send the next units in the same format.

continue

Good—let's continue in the same **20 questions + clear descriptive answers** format so you can revise smoothly.

SUBJECT 1: Information & Cyber Security

Unit III: Public Key & Key Management (20 Questions)

1. What is Public Key Cryptography?

Public key cryptography is a method of encryption that uses two keys: a public key (shared) and a private key (kept secret). It ensures secure communication over insecure networks.

2. What is the RSA Algorithm?

RSA is a widely used public-key algorithm based on the mathematical difficulty of factoring large prime numbers. It is used for encryption and digital signatures.

3. How does RSA work?

- Generate two large prime numbers
 - Compute public and private keys
 - Encrypt using public key
 - Decrypt using private key
-

4. What is Key Management?

Key management involves generating, distributing, storing, and protecting cryptographic keys.

5. What is Key Distribution?

It is the process of securely sharing cryptographic keys between sender and receiver.

6. What is Diffie-Hellman Key Exchange?

A method that allows two parties to securely share a secret key over an insecure channel.

7. What is a Digital Signature?

A digital signature is a cryptographic technique used to verify authenticity and integrity of a message.

8. How are Digital Signatures created?

- Generate hash of message
 - Encrypt hash with sender's private key
-

9. What is Message Digest?

A fixed-size hash value representing data, used to ensure integrity.

10. What is a Hash Function?

A function that converts input data into a fixed-length output.

11. What is Authentication?

Process of verifying identity of a user or system.

12. What is X.509 Certificate?

A digital certificate that verifies identity in networks using public key infrastructure (PKI).

13. What is PKI (Public Key Infrastructure)?

A framework for managing digital certificates and public-key encryption.

14. What is Key Length?

Size of the cryptographic key; longer keys provide stronger security.

15. What is Key Escrow?

A system where keys are stored by a trusted third party.

16. What is Digital Certificate Authority (CA)?

An entity that issues digital certificates.

17. What is Encryption vs Digital Signature?

Encryption ensures confidentiality, while digital signature ensures authenticity and integrity.

18. What is Non-repudiation in PKI?

Ensures that a sender cannot deny sending a message.

19. What are Applications of Public Key Cryptography?

- Secure email
 - Online banking
 - Digital signatures
-

20. What are Limitations of Public Key Cryptography?

- Slower than symmetric encryption
 - Requires complex key management
-
-

◆ Unit IV: Security Requirements (20 Questions)

1. What is IP Security (IPSec)?

A protocol suite used to secure IP communications through encryption and authentication.

2. Modes of IPSec

- Transport Mode
 - Tunnel Mode
-

3. What is SSL?

Secure Sockets Layer is a protocol used to secure internet communication.

4. What is TLS?

Transport Layer Security is the modern version of SSL.

5. What is HTTPS?

Secure HTTP that uses SSL/TLS for encrypted communication.

6. What is Secure Email?

Email protection using encryption and authentication techniques.

7. What is PGP?

Pretty Good Privacy is used for secure email communication.

8. What is S/MIME?

Secure/Multipurpose Internet Mail Extensions used for email encryption.

9. What is SET?

Secure Electronic Transaction protocol used for secure online payments.

10. What is S-HTTP?

Secure version of HTTP providing message-level security.

11. What is Email Spoofing?

Forging email headers to appear as trusted source.

12. What is Phishing Email?

Fake email designed to steal sensitive information.

13. What is Secure Socket Layer Architecture?

Includes handshake protocol, record protocol, and alert protocol.

14. What is Encryption in Network Security?

Process of securing data during transmission.

15. What is Digital Envelope?

Combines symmetric and asymmetric encryption.

16. What is Authentication in Email?

Verifying sender identity using digital signatures.

17. What is Data Integrity in Communication?

Ensuring data is not modified during transmission.

18. What is VPN?

Virtual Private Network creates secure communication over public network.

19. What are Security Protocols?

Rules that ensure secure data exchange.

20. What are Limitations of Security Protocols?

- Performance overhead
 - Complex implementation
-
-

◆ Unit V: Firewall & Intrusion (20 Questions)

1. What is a Firewall?

A firewall is a network security system that monitors and controls traffic based on predefined rules.

2. Types of Firewalls

- Packet Filtering
 - Stateful Inspection
 - Proxy Firewall
-

3. What is Packet Filtering Firewall?

Filters packets based on IP, port, and protocol.

4. What is Stateful Firewall?

Tracks active connections and makes decisions accordingly.

5. What is Proxy Firewall?

Acts as an intermediary between users and internet.

6. What is Intrusion Detection System (IDS)?

Detects suspicious activities in a network.

7. What is Intrusion Prevention System (IPS)?

Detects and prevents attacks.

8. IDS Types

- Signature-based
 - Anomaly-based
-

9. What is DMZ?

A network zone that separates internal and external networks.

10. What is Access Control?

Restricting access to resources.

11. What is Firewall Architecture?

Design of firewall deployment in network.

12. What is Threat Detection?

Identifying malicious activities.

13. What is False Positive in IDS?

Normal activity detected as attack.

14. What is False Negative?

Attack not detected.

15. What are Firewall Limitations?

Cannot stop insider threats or encrypted attacks fully.

16. What is Network Address Translation (NAT)?

Hides internal IP addresses.

17. What is Bastion Host?

A highly secured system exposed to internet.

18. What is Access Control List (ACL)?

Rules defining permitted traffic.

19. What is Logging in Firewall?

Recording traffic activity.

20. What are Challenges in Firewall Systems?

- Complexity
 - Performance issues
-
-

◆ Unit VI: Cyber Forensics, Hacking & Countermeasures (20 Questions)

1. What is Cyber Forensics?

Process of collecting and analyzing digital evidence.

2. Steps in Cyber Forensics

- Identification
- Collection
- Analysis

- Reporting
-

3. What is Ethical Hacking?

Legal hacking to find vulnerabilities.

4. Types of Hackers

- White hat
 - Black hat
 - Grey hat
-

5. What is Malware?

Malicious software like virus, worm, trojan.

6. What is Virus?

Self-replicating malicious program.

7. What is Worm?

Spreads without user action.

8. What is Trojan Horse?

Malware disguised as legitimate software.

9. What is Phishing?

Fraudulent attempt to steal data.

10. What is DoS Attack?

Denial of Service attack making system unavailable.

11. What is DDoS Attack?

Distributed attack using multiple systems.

12. What is Spyware?

Software that secretly monitors users.

13. What is Ransomware?

Locks data and demands payment.

14. What is Digital Evidence?

Data used in investigation.

15. What is Chain of Custody?

Tracking evidence handling process.

16. What is Privacy Protection?

Safeguarding personal data.

17. What is Cyber Law (India)?

Legal framework for cyber crimes.

18. What are Countermeasures?

Actions taken to prevent attacks.

19. What is Security Awareness?

Educating users about threats.

20. What are Challenges in Cyber Forensics?

- Data volume
 - Encryption
 - Legal issues
-
-

 **Done: FULL Cyber Security (All Units)**

SUBJECT 2: Cloud Computing

Unit I: Introduction to Cloud Computing (20 Questions)

1. What is Cloud Computing?

Cloud computing is the delivery of computing services (servers, storage, databases, networking, software) over the internet on a pay-as-you-go basis.

2. What are the key characteristics of Cloud Computing?

- On-demand self-service
 - Broad network access
 - Resource pooling
 - Rapid elasticity
 - Measured service
-

3. What is On-demand Self-service?

Users can access computing resources automatically without human interaction.

4. What is Resource Pooling?

Cloud providers serve multiple users using shared resources.

5. What is Rapid Elasticity?

Resources can be scaled up or down quickly as needed.

6. What is Measured Service?

Users pay only for what they use.

7. What are Cloud Service Models?

- SaaS
 - PaaS
 - IaaS
-

8. What is SaaS?

Software delivered over the internet (e.g., email services).

9. What is PaaS?

Platform for developing and deploying applications.

10. What is IaaS?

Provides virtualized computing resources like servers and storage.

11. What are Deployment Models?

- Public Cloud
 - Private Cloud
 - Hybrid Cloud
-

12. What is Public Cloud?

Services offered over the internet to multiple users.

13. What is Private Cloud?

Cloud used by a single organization.

14. What is Hybrid Cloud?

Combination of public and private clouds.

15. What is Cloud Migration?

Process of moving data/applications to cloud.

16. What is Cloud Architecture?

Structure of cloud system including front-end and back-end.

17. What is Utility Computing?

Pay-per-use computing model.

18. What are Benefits of Cloud?

- Cost saving
 - Scalability
 - Flexibility
-

19. What are Challenges in Cloud?

- Security issues
 - Downtime
 - Vendor lock-in
-

20. What is Cloud Deployment Model Selection?

Choosing cloud type based on business needs.

◆ **Unit II: Data Storage and Cloud Computing (20 Questions)**

1. What is Cloud Storage?

Storing data on remote servers accessed via internet.

2. Types of Cloud Storage

- Object storage
 - Block storage
 - File storage
-

3. What is Object Storage?

Stores data as objects with metadata.

4. What is Block Storage?

Stores data in fixed-size blocks.

5. What is File Storage?

Stores data in hierarchical file system.

6. What is Data Management?

Process of storing, organizing, and retrieving data.

7. What is Data Backup?

Creating copies of data to prevent loss.

8. What is Data Replication?

Storing copies of data across multiple locations.

9. What is Data Consistency?

Ensuring all users see the same data.

10. What is Storage Virtualization?

Combining multiple storage devices into one system.

11. What is Distributed Storage?

Data stored across multiple machines.

12. What is Cloud Database?

Database hosted in cloud.

13. What is Data Lifecycle?

Stages: creation → storage → usage → deletion.

14. What is Data Security in Cloud?

Protection using encryption and access control.

15. What is Data Availability?

Ensuring data is accessible anytime.

16. What is Data Integrity?

Ensuring data is accurate and unchanged.

17. What is Data Compression?

Reducing data size to save storage.

18. What is Data Deduplication?

Removing duplicate data.

19. What is Big Data in Cloud?

Handling large-scale data using cloud resources.

20. What are Storage Challenges?

- Latency
 - Security risks
 - Cost
-
-

◆ Unit III: Virtualization in Cloud Computing (20 Questions)

1. What is Virtualization?

Creating virtual versions of hardware or software resources.

2. What is a Hypervisor?

Software that creates and manages virtual machines.

3. Types of Hypervisors

- Type 1 (bare-metal)
 - Type 2 (hosted)
-

4. What is Full Virtualization?

Complete simulation of hardware.

5. What is Partial Virtualization?

Only part of hardware is virtualized.

6. What is Server Virtualization?

Running multiple servers on one physical machine.

7. What is Storage Virtualization?

Combining multiple storage devices.

8. What is Network Virtualization?

Creating virtual networks.

9. What is Desktop Virtualization?

Running desktop environments on servers.

10. What is Application Virtualization?

Running applications without installation.

11. What is Grid Computing?

Distributed computing across multiple systems.

12. What is Cluster Computing?

Multiple systems working as one.

13. What is Virtual Machine (VM)?

Software-based computer system.

14. What is Containerization?

Lightweight virtualization (e.g., Docker).

15. What are Advantages of Virtualization?

- Better resource utilization
 - Cost reduction
-

16. What are Limitations?

- Performance overhead
 - Complexity
-

17. What is Live Migration?

Moving VM without downtime.

18. What is Load Balancing?

Distributing workload across systems.

19. What is Virtual Network?

Software-defined network.

20. What is CPU Virtualization?

Sharing CPU among multiple VMs.

◆ Unit IV: Cloud Platforms and Applications (20 Questions)

1. What is AWS?

Cloud platform providing computing services.

2. What is Microsoft Azure?

Cloud platform by Microsoft.

3. What is Google Cloud?

Cloud services by Google.

4. What is Amazon EC2?

Virtual server service.

5. What is Amazon S3?

Object storage service.

6. What is Cloud Application?

Application hosted on cloud.

7. What is OpenStack?

Open-source cloud platform.

8. What is Cloud in Healthcare?

Data storage and remote patient monitoring.

9. What is Cloud in Education?

Online learning platforms.

10. What is Cloud in Business?

ERP, CRM systems.

11. What is Cloud Gaming?

Games hosted on cloud servers.

12. What is Cloud ERP?

Enterprise resource planning via cloud.

13. What is SaaS Application Example?

Google Docs, Gmail.

14. What is PaaS Example?

Google App Engine.

15. What is IaaS Example?

AWS EC2.

16. What is Multi-tenant Architecture?

Multiple users share same infrastructure.

17. What is Scalability in Applications?

Ability to handle increased load.

18. What is Fault Tolerance?

System continues even after failure.

19. What is API in Cloud?

Interface for communication between services.

20. What are Cloud Application Benefits?

- Accessibility

- Cost efficiency
-
-

◆ Unit V: Security in Cloud Computing (20 Questions)

1. What are Risks in Cloud?

Data breach, loss, unauthorized access.

2. What is Data Security?

Protecting data from threats.

3. What is Identity Management?

Managing user identities.

4. What is Access Control?

Restricting access to resources.

5. What is Multi-factor Authentication?

Using multiple verification methods.

6. What is Encryption in Cloud?

Protecting data using cryptography.

7. What is Data Privacy?

Protecting personal data.

8. What is Insider Threat?

Threat from internal users.

9. What is Cloud Vulnerability?

Weakness in cloud system.

10. What is Secure API?

Protected communication interface.

11. What is Cloud Compliance?

Following legal standards.

12. What is Data Leakage?

Unauthorized data exposure.

13. What is Backup in Cloud?

Copy of data for recovery.

14. What is Disaster Recovery?

Restoring data after failure.

15. What is Secure Development?

Developing secure applications.

16. What is Cloud Security Testing?

Testing for vulnerabilities.

17. What is Availability in Cloud?

Ensuring uptime.

18. What is Security SLA?

Agreement on security standards.

19. What are Cloud Security Challenges?

- Multi-tenancy
 - Data loss
-

20. What is Auditing in Cloud?

Monitoring and reviewing security.

◆ Unit VI: Advanced Cloud Topics (20 Questions)

1. What is Edge Computing?

Processing data near source.

2. What is Fog Computing?

Intermediate layer between cloud and devices.

3. What is Serverless Computing?

Running code without managing servers.

4. What is Docker?

Containerization platform.

5. What is Kubernetes?

Manages containerized applications.

6. What is DevOps?

Integration of development and operations.

7. What is CI/CD?

Continuous integration and delivery.

8. What is Microservices Architecture?

Application divided into small services.

9. What is IoT in Cloud?

Connecting devices to cloud.

10. What is Big Data in Cloud?

Processing large datasets.

11. What is Cloud Automation?

Automating cloud operations.

12. What is Cloud Monitoring?

Tracking performance.

13. What is Cloud Orchestration?

Managing multiple cloud services.

14. What is Multi-cloud?

Using multiple cloud providers.

15. What is Hybrid IT?

Mix of cloud and on-premise.

16. What is GPU Computing in Cloud?

Using GPU for high performance tasks.

17. What is AI in Cloud?

Using cloud for machine learning.

18. What is Blockchain in Cloud?

Distributed ledger using cloud.

19. What is Cloud-native Application?

Designed for cloud environment.

20. What are Future Trends in Cloud?

- AI integration
 - Serverless growth
 - Edge computing
-

 **DONE: FULL SYLLABUS (Both Subjects)**

1. RSA Algorithm (Detailed Explanation)

◆ What is RSA?

RSA is a public key cryptography algorithm used for:

- Secure data transmission
- Encryption & decryption
- Digital signatures

It uses two keys:

- Public Key → for encryption
 - Private Key → for decryption
-

♦ Why RSA is Important?

RSA solves the major problem of secure key sharing.

Even if someone intercepts the message, they cannot decrypt it without the private key.

♦ Working of RSA (Step-by-Step)

✅ Step 1: Choose Two Prime Numbers

Select two large prime numbers:

- p and q

Example:

$p = 3, q = 11$

✅ Step 2: Compute n

$$n = p \times q$$

Example:

$n = 3 \times 11 = 33$

✅ Step 3: Compute Euler's Totient Function

$$\phi(n) = (p - 1)(q - 1)$$

Example:

$\phi(n) = (3-1)(11-1) = 2 \times 10 = 20$

✅ Step 4: Choose Public Key (e)

Choose e such that:

- $1 < e < \phi(n)$
- $\gcd(e, \phi(n)) = 1$

Example:

$e = 3$

✓ Step 5: Compute Private Key (d)

Find d such that:

$$d \times e \bmod \phi(n) = 1$$

Example:

$d = 7$ (because $7 \times 3 = 21 \equiv 1 \bmod 20$)

◆ Keys Generated

- Public Key $\rightarrow (e, n) = (3, 33)$
 - Private Key $\rightarrow (d, n) = (7, 33)$
-

◆ Encryption Process

$$C = M^e \bmod n$$

Where:

- M = message
- C = ciphertext

Example:

$M = 4$

$C = 4^3 \bmod 33 = 64 \bmod 33 = 31$

◆ Decryption Process

$$M = C^d \bmod n$$

Example:

$$M = 31^7 \bmod 33 = 4$$

◆ Diagram (Concept)

Sender → Encrypt (Public Key) → Ciphertext → Receiver → Decrypt (Private Key)

◆ Advantages of RSA

- Secure key exchange
 - Widely used (HTTPS, banking)
 - Supports digital signatures
-

◆ Disadvantages

- Slow compared to symmetric encryption
 - Requires large key size
 - Computationally expensive
-

◆ Applications of RSA

- Secure web communication (HTTPS)
 - Email encryption
 - Digital signatures
-

1. DES (Data Encryption Standard)

◆ What is DES?

DES is a **symmetric key encryption algorithm** that uses the **same key for encryption and decryption**. It was one of the earliest widely used encryption standards.

◆ Key Features of DES

- Key size: **56 bits**
 - Block size: **64 bits**
 - Type: **Block cipher**
 - Developed by IBM
-

♦ Working of DES (Step-by-Step)

✓ 1. Input Block

- Takes **64-bit plaintext**
-

✓ 2. Initial Permutation

- Rearranges bits according to a fixed pattern
-

✓ 3. Splitting

- Divides data into two halves:
 - Left (L)
 - Right (R)
-

✓ 4. 16 Rounds of Processing

Each round includes:

- Expansion of right half
 - XOR with round key
 - Substitution using S-boxes
 - Permutation
-

✓ 5. Swapping

- Left and right halves are swapped
-

✅ 6. Final Permutation

- Produces final ciphertext
-

♦ Why DES is Weak Today?

- Small key size (56-bit)
 - Vulnerable to **brute-force attacks**
 - Can be broken with modern computers
-

♦ Advantages of DES

- Simple structure
 - Faster in hardware
-

♦ Disadvantages

- Not secure anymore
 - Easily breakable
-

♦ Viva Tip (DES in one line)

“DES is a symmetric block cipher using a 56-bit key and 16 rounds, but it is now considered insecure due to its small key size.”

🔒 2. AES (Advanced Encryption Standard)

♦ What is AES?

AES is a **modern symmetric encryption algorithm** used worldwide for secure data encryption.

♦ Key Features of AES

- Key sizes: **128, 192, 256 bits**

- Block size: **128 bits**
 - More secure and faster than DES
-

◆ Working of AES

✓ 1. Input Block

- Takes **128-bit plaintext**
-

✓ 2. Initial Round

- AddRoundKey (XOR with key)
-

✓ 3. Multiple Rounds (10, 12, or 14)

Depending on key size:

- 10 rounds → 128-bit key
- 12 rounds → 192-bit key
- 14 rounds → 256-bit key

Each round includes:

- SubBytes (substitution)
 - ShiftRows (row shifting)
 - MixColumns (mixing data)
 - AddRoundKey
-

✓ 4. Final Round

- Similar but without MixColumns
-

◆ Why AES is Strong?

- Large key size
- Complex structure

- Resistant to brute-force attacks
-

◆ Advantages of AES

- Highly secure
 - Fast in software and hardware
 - Widely used
-

◆ Disadvantages

- More complex than DES
 - Slightly higher computational requirement
-

◆ Applications of AES

- Banking systems
 - Wi-Fi security
 - Government data protection
-

◆ Viva Tip (AES in one line)

“AES is a symmetric block cipher with key sizes of 128, 192, or 256 bits, providing strong security and widely used for modern encryption.”

3. Difference Between AES and DES

Feature	DES	AES
Key Size	56-bit	128/192/256-bit
Block Size	64-bit	128-bit
Security	Weak	Strong
Rounds	16	10/12/14
Speed	Slower (software)	Faster

Feature	DES	AES
Structure	Feistel	Substitution-Permutation
Usage	Outdated	Widely used

1. Firewall

◆ What is a Firewall?

A firewall is a **network security system** that monitors and controls incoming and outgoing traffic based on predefined security rules.

👉 It acts like a **gatekeeper** between:

- Internal network (trusted)
 - External network (untrusted, like the internet)
-

◆ How Firewall Works?

- Checks data packets entering/leaving network
 - Applies rules (allow/deny)
 - Blocks unauthorized access
-

◆ Types of Firewalls

1. Packet Filtering Firewall

- Filters packets based on:
 - IP address
 - Port number
 - Fast but less secure
-

2. Stateful Inspection Firewall

- Tracks active connections

- Makes smarter decisions
 - More secure than packet filtering
-

3. Proxy Firewall

- Acts as an intermediary
 - Hides internal network
 - Provides high security
-

♦ Advantages of Firewall

- Prevents unauthorized access
 - Protects network from attacks
 - Controls traffic
-

♦ Limitations of Firewall

- Cannot stop insider attacks
 - Cannot detect encrypted threats fully
 - Needs proper configuration
-

♦ Viva Tip (Firewall in one line)

“A firewall is a security system that monitors and controls network traffic based on predefined rules to protect systems from unauthorized access.”

2. IDS (Intrusion Detection System)

♦ What is IDS?

IDS is a system that **detects suspicious or malicious activities** in a network.

 It only **detects**, not blocks.

◆ How IDS Works?

- Monitors network traffic
 - Compares with known attack patterns
 - Alerts administrator if threat detected
-

◆ Types of IDS

1. Signature-Based IDS

- Detects known attacks
 - Uses database of attack patterns
-

2. Anomaly-Based IDS

- Detects unusual behavior
 - Can find new attacks
-

◆ Advantages of IDS

- Detects attacks early
 - Provides alerts
 - Helps in monitoring
-

◆ Disadvantages

- Cannot stop attacks
 - May give false alarms
-

◆ Viva Tip (IDS in one line)

“IDS is a system that monitors network traffic and detects suspicious activities but does not prevent them.”

3. IPS (Intrusion Prevention System)

◆ What is IPS?

IPS is a system that **detects AND prevents attacks** in real-time.

👉 It is an advanced version of IDS.

◆ How IPS Works?

- Monitors traffic
 - Detects threats
 - Automatically blocks malicious activity
-

◆ Types of IPS

- Network-based IPS
 - Host-based IPS
-

◆ Advantages of IPS

- Prevents attacks automatically
 - Real-time protection
 - Reduces manual work
-

◆ Disadvantages

- Can block legitimate traffic (false positives)
 - More complex
-

◆ Viva Tip (IPS in one line)

“IPS is a security system that detects and actively blocks malicious activities in a network.”

⚖️ **Difference: Firewall vs IDS vs IPS**

Feature	Firewall	IDS	IPS
Function	Filters traffic	Detects attacks	Detects + Prevents
Action	Allow/deny	Alert only	Block + alert
Position	Network boundary	Inside network	Inline with traffic
Response	Passive	Passive	Active

🔥 Final Viva Answer (Best Combined Answer)

If examiner asks:

👉 “Explain Firewall, IDS, IPS”

Say:

“A firewall controls network traffic based on rules to prevent unauthorized access. IDS monitors and detects suspicious activities but does not take action, while IPS not only detects but also prevents attacks in real-time by blocking malicious traffic.”

🎯 Easy Way to Remember

- Firewall → **Gatekeeper**
- IDS → **Alarm system**
- IPS → **Security guard (takes action)**

💡 Virtualization (Detailed Explanation)

◆ What is Virtualization?

Virtualization is the process of creating a **virtual (software-based) version of physical resources** like:

- Servers
- Storage
- Networks
- Operating systems

👉 In simple words:

One physical machine can run multiple virtual machines (VMs).

◆ Why Virtualization is Needed?

Before virtualization:

- One server = one application (waste of resources)

After virtualization:

- One server = multiple applications
 - Better **resource utilization** and **cost saving**
-

◆ What is a Virtual Machine (VM)?

A Virtual Machine is a **software-based computer** that behaves like a real computer.

Each VM has:

- OS
 - CPU
 - Memory
 - Storage
-

◆ What is a Hypervisor?

A **hypervisor** is software that creates and manages virtual machines.

◆ Types of Hypervisor

✅ Type 1 (Bare Metal)

- Runs directly on hardware
 - More efficient and secure
 - Example: used in data centers
-

✅ Type 2 (Hosted)

- Runs on an operating system
 - Easier to use
 - Example: VirtualBox
-

◆ Types of Virtualization

1. Server Virtualization

- Multiple virtual servers on one physical server
-

2. Storage Virtualization

- Combines multiple storage devices into one
-

3. Network Virtualization

- Creates virtual networks
-

4. Desktop Virtualization

- Desktop runs on remote server
-

5. Application Virtualization

- Apps run without installation
-

◆ How Virtualization Works (Simple Flow)

Physical Hardware → Hypervisor → Multiple VMs → Each VM runs OS & apps

◆ Advantages of Virtualization

- Better resource utilization
- Cost reduction
- Easy backup and recovery

- Scalability
 - Isolation between VMs
-

◆ Disadvantages

- Performance overhead
 - Complex management
 - Security risks if misconfigured
-

◆ What is Live Migration?

Moving a virtual machine from one server to another **without downtime**.

◆ What is Load Balancing?

Distributing workload across multiple systems to improve performance.

◆ Virtualization vs Cloud (Important Viva Question)

- Virtualization → Technology
- Cloud → Service using virtualization

👉 Cloud uses virtualization to provide services over internet.

◆ Viva Tip (Virtualization in one line)

“Virtualization is the creation of virtual versions of computing resources using a hypervisor, allowing multiple virtual machines to run on a single physical system.”

🔥 Common Viva Questions from this Topic

- What is hypervisor?
- Difference between Type 1 and Type 2
- What is VM?
- Benefits of virtualization

- Virtualization vs cloud
-

Easy Example (Use in Viva)

“Like one physical building divided into multiple apartments, virtualization divides one physical server into multiple virtual machines.”

◆ What are Cloud Service Models?

Cloud service models define **what type of services are provided by the cloud** and **how much control the user has**.

👉 There are **3 main models**:

- IaaS (Infrastructure as a Service)
 - PaaS (Platform as a Service)
 - SaaS (Software as a Service)
-

1. IaaS (Infrastructure as a Service)

◆ What is IaaS?

IaaS provides **basic computing resources** like:

- Virtual machines
- Storage
- Networks

👉 User has **maximum control**

◆ What user manages?

- OS
- Applications
- Data

Cloud provider manages:

- Hardware
 - Networking
-

◆ Examples

- Amazon EC2
 - Microsoft Azure VMs
-

◆ Advantages

- High flexibility
 - Full control
 - Scalable
-

◆ Disadvantages

- Requires technical knowledge
 - User responsible for security
-

◆ Viva Tip (IaaS in one line)

“IaaS provides virtualized hardware resources where users manage OS and applications.”

2. PaaS (Platform as a Service)

◆ What is PaaS?

PaaS provides a **platform for developers** to build, test, and deploy applications.

◆ What user manages?

- Applications
- Data

Cloud provider manages:

- OS
 - Runtime
 - Infrastructure
-

◆ Examples

- Google App Engine
 - Microsoft Azure App Services
-

◆ Advantages

- Easy development
 - No need to manage OS
 - Faster deployment
-

◆ Disadvantages

- Less control
 - Vendor lock-in
-

◆ Viva Tip (PaaS in one line)

“PaaS provides a platform for application development without managing underlying infrastructure.”

3. SaaS (Software as a Service)

◆ What is SaaS?

SaaS provides **ready-to-use software** over the internet.

👉 Users just use the software—no installation needed.

◆ What user manages?

- Only usage

Cloud provider manages:

- Everything (app, OS, infrastructure)
-

◆ Examples

- Gmail
 - Google Docs
-

◆ Advantages

- Easy to use
 - No installation
 - Accessible anywhere
-

◆ Disadvantages

- Limited control
 - Requires internet
-

◆ Viva Tip (SaaS in one line)

“SaaS provides software applications over the internet where users only access the service without managing anything.”

Difference Between IaaS, PaaS, SaaS

Feature	IaaS	PaaS	SaaS
Control	High	Medium	Low
User manages	OS + Apps	Apps only	Nothing
Provider manages	Hardware	OS + Hardware	Everything

Feature	IaaS	PaaS	SaaS
Usage	IT admins	Developers	End users

Best Viva Answer (Very Important)

If examiner asks:

 **“Explain SaaS, PaaS, IaaS”**

Say:

“Cloud service models include IaaS, PaaS, and SaaS. IaaS provides infrastructure like virtual machines where users manage OS and applications. PaaS provides a platform for developers to build applications without managing infrastructure. SaaS delivers software over the internet where users simply use the application without managing anything.”

Easy Trick to Remember

- IaaS → “I build everything”
- PaaS → “I build apps only”
- SaaS → “I just use software”

  +  **1-Page Revision Sheet (Cloud + Cyber Security)**

Cyber Security Key Points

CIA Triad

- **Confidentiality** → data access restricted
 - **Integrity** → data not modified
 - **Availability** → data always accessible
-

Cryptography

- Encryption → plaintext → ciphertext

- Decryption → ciphertext → plaintext
 - Symmetric → same key
 - Asymmetric → public + private key
-

Algorithms

- **DES** → 56-bit key (weak, outdated)
 - **AES** → 128/192/256-bit (strong, modern)
 - **RSA** → public key encryption (secure communication)
-

Security Tools

- **Firewall** → blocks unauthorized access
 - **IDS** → detects attacks
 - **IPS** → detects + blocks attacks
-

Attacks

- Malware → virus, worm, trojan
 - Phishing → fake emails
 - DoS/DDoS → system overload
-

Virtualization

- One physical machine → multiple VMs
 - Hypervisor → manages VMs
 - Type 1 → bare metal
 - Type 2 → hosted
-
-

Cloud Computing Key Points

Characteristics

- On-demand
 - Scalable
 - Pay-as-you-use
 - Resource pooling
-

Service Models

- **IaaS** → infrastructure (EC2)
 - **PaaS** → platform (App Engine)
 - **SaaS** → software (Gmail)
-

Deployment Models

- Public Cloud
 - Private Cloud
 - Hybrid Cloud
-

Storage

- Object / Block / File storage
 - Backup + replication for safety
-

Advanced Topics

- Docker → containers
 - Kubernetes → container management
 - Edge computing → near data processing
 - Serverless → no server management
-
-

Top 30 Expected Viva Questions (MOST IMPORTANT)

Cyber Security

1. What is Information Security?
 2. Explain CIA Triad.
 3. What is cryptography?
 4. Difference between symmetric and asymmetric encryption?
 5. What is RSA algorithm?
 6. What is AES?
 7. Why is DES not secure?
 8. What is encryption and decryption?
 9. What is a digital signature?
 10. What is a hash function?
 11. What is a firewall?
 12. Types of firewall?
 13. What is IDS?
 14. What is IPS?
 15. Difference between IDS and IPS?
-

Cloud Computing

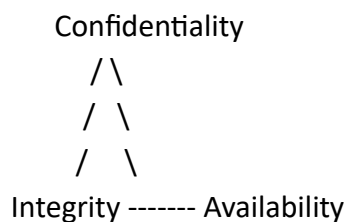
16. What is cloud computing?
17. Characteristics of cloud computing?
18. What are SaaS, PaaS, IaaS?
19. Difference between SaaS, PaaS, IaaS?
20. What is virtualization?
21. What is a hypervisor?
22. Types of hypervisor?
23. What is cloud storage?
24. Types of cloud deployment models?
25. What is cloud migration?

- 26. What is AWS?
- 27. What is Azure?
- 28. What is Google Cloud?
- 29. What is Docker?
- 30. What is Kubernetes?

CYBER SECURITY – QUICK DIAGRAMS + TRICKS

1. CIA Triad (VERY IMPORTANT)

Diagram (draw triangle)



Memory Trick:

“CIA = Keep Data Safe Always”

- C = Confidential (hide data)
- I = Integrity (no change)
- A = Availability (always accessible)

2. Symmetric vs Asymmetric Encryption

Diagram

Symmetric:

Sender ---- Same Key ---- Receiver

Asymmetric:

Sender -- Public Key --> Receiver

Receiver -- Private Key --> Decrypt

🧠 Trick:

👉 “Symmetric = Same, Asymmetric = Different”

3. RSA Flow Diagram

Plaintext

↓ (Encrypt with Public Key)

Ciphertext

↓ (Decrypt with Private Key)

Plaintext

🧠 Trick:

👉 “Public locks, Private unlocks”

4. Firewall / IDS / IPS

Internet

↓

[Firewall] → filters traffic

↓

[IDS] → detects attacks

↓

[IPS] → blocks attacks

↓

Internal Network

🧠 Trick:

👉 “F-D-B”

- Firewall = Filter
 - IDS = Detect
 - IPS = Block
-

5. Types of Attacks

Malware → Virus / Worm / Trojan

Phishing → Fake emails

DoS → System overload

 **Trick:**

 **“M-P-D = Make People Down”**

CLOUD COMPUTING – QUICK DIAGRAMS + TRICKS

1. Cloud Service Models (MOST IMPORTANT)

SaaS → Software (Gmail)

PaaS → Platform (App Engine)

IaaS → Infrastructure (EC2)

 **Ladder Diagram**

SaaS (User uses app)



PaaS (Build apps)



IaaS (Manage servers)

 **Trick:**

 **“S-P-I = Software, Platform, Infrastructure”**

 Easy memory:

- SaaS → Use
 - PaaS → Build
 - IaaS → Control
-

2. Deployment Models

Public Cloud → Everyone

Private Cloud → One company

Hybrid Cloud → Both

 **Trick:**

 **“PPH = Public, Private, Hybrid”**

3. Virtualization Diagram

Physical Server



Hypervisor



VM1 VM2 VM3

🧠 Trick:

👉 “One machine, many minds (VMs)”

4. Hypervisor Types

Type 1 → Bare Metal (fast)

Type 2 → Hosted (easy)

🧠 Trick:

👉 “1 = Strong, 2 = Simple”

5. Cloud Architecture Flow

User



Internet



Cloud Provider



Services (SaaS / PaaS / IaaS)

🧠 Trick:

👉 “User → Net → Cloud → Service”

⚡ FINAL SUPER QUICK REVISION TRICKS

🔒 Security

- CIA → Safe Data Triangle
- RSA → Public lock, Private key

- AES → Strong modern lock
 - DES → Old weak lock
 - Firewall → Gatekeeper
 - IDS → Alarm
 - IPS → Security guard
-

Cloud

- SaaS → Use apps
- PaaS → Build apps
- IaaS → Control machines
- Virtualization → One → Many VMs
- Cloud → Internet-based computing

CLOUD COMPUTING – IMPORTANT TOPICS (FROM YOUR SYLLABUS)

◆ UNIT I: INTRODUCTION TO CLOUD COMPUTING

1. What is Cloud Computing (MOST IMPORTANT)

Cloud computing is a technology that provides **computing services like storage, servers, databases, networking, and software over the internet.**

 Instead of owning physical hardware, users **rent resources from cloud providers.**

✓ **Key idea:**

“Use computing resources like electricity—pay only for what you use.”

2. Characteristics of Cloud Computing

Important features:

On-demand self service

Users can access resources automatically without human interaction.

Broad network access

Services are available via internet on any device.

Resource pooling

Provider shares resources among multiple users.

Rapid elasticity

Resources can scale up/down quickly.

Measured service

Pay-as-you-use model.

3. Service Models (VERY IMPORTANT)

IaaS (Infrastructure as a Service)

- Provides virtual machines, storage, networks
- User manages OS and applications
- Example: AWS EC2

 “Rent hardware in cloud”

PaaS (Platform as a Service)

- Provides platform for development
- User only writes code
- Example: Google App Engine

 “Build apps without managing OS”

SaaS (Software as a Service)

- Ready-to-use software
- No installation needed
- Example: Gmail, Google Docs

 “Just use software”

4. Deployment Models

Public Cloud

- Open to all users
- Example: AWS, Azure

Private Cloud

- Used by one organization only

Hybrid Cloud

- Combination of public + private

👉 Trick:

Public = everyone, Private = one, Hybrid = both

5. Cloud Architecture (Basic Idea)

Cloud has 3 layers:

- Frontend (user interface)
 - Backend (servers, storage)
 - Network (internet connection)
-

6. Benefits of Cloud Computing

- Cost saving
 - Scalability
 - Easy access
 - Backup & recovery
 - No hardware maintenance
-

7. Limitations

- Internet dependency
- Security concerns

- Vendor lock-in
-

◆ UNIT II: DATA STORAGE IN CLOUD

1. Cloud Storage

Cloud storage means **storing data on remote servers accessed via internet.**

2. Types of Cloud Storage

Object Storage

- Stores data as objects
- Used for images, videos

Block Storage

- Data stored in blocks
- Used for databases

File Storage

- Stores data in folders
 - Like normal file system
-

3. Data Management

Includes:

- Storage
 - Backup
 - Retrieval
 - Security
-

4. Data Replication

Copies of data are stored in multiple locations to ensure **availability and backup.**

5. Data Security

Includes:

- Encryption
 - Access control
 - Authentication
-

6. Big Data in Cloud

Cloud helps process **large datasets quickly** using distributed systems.

◆ UNIT III: VIRTUALIZATION

1. Virtualization (VERY IMPORTANT)

Virtualization is the process of creating **multiple virtual machines (VMs) on a single physical system**.

👉 One computer behaves like many computers.

2. Hypervisor

A hypervisor is software that:

- Creates VMs
 - Manages resources
-

3. Types of Hypervisor

Type 1 (Bare metal)

- Runs directly on hardware
- Fast and secure

Type 2 (Hosted)

- Runs on OS
 - Easy to use
-

4. Types of Virtualization

- Server virtualization
 - Storage virtualization
 - Network virtualization
 - Desktop virtualization
-

5. Advantages

- Resource efficiency
 - Cost saving
 - Isolation
 - Scalability
-
-

◆ UNIT IV: CLOUD PLATFORMS

1. AWS (Amazon Web Services)

Most popular cloud platform providing:

- Computing
 - Storage
 - Databases
-

2. Microsoft Azure

Cloud platform by Microsoft used for enterprise solutions.

3. Google Cloud Platform

Used for AI, data analytics, and machine learning.

4. Cloud Applications

Examples:

- Gmail (SaaS)
 - Google Drive
 - Dropbox
-

5. Cloud in Real Life

- Education → online classes
 - Healthcare → patient data
 - Business → ERP systems
-
-

◆ UNIT V: SECURITY IN CLOUD

1. Cloud Security

Protection of data, applications, and infrastructure in cloud.

2. Identity Management

Ensures only authorized users can access cloud resources.

3. Access Control

Controls who can access what resources.

4. Encryption

Data is converted into unreadable form to protect it.

5. Cloud Risks

- Data breach
 - Data loss
 - Unauthorized access
-

6. Disaster Recovery

Restoring data after system failure.

◆ UNIT VI: ADVANCED CLOUD TOPICS

1. Edge Computing

Data is processed **near the source** instead of central cloud.

👉 Reduces delay (latency)

2. Serverless Computing

Developers run code without managing servers.

👉 Cloud automatically handles everything.

3. Docker (Containers)

Tool that packages applications with all dependencies.

👉 Lightweight virtualization

4. Kubernetes

Manages and controls multiple containers.

5. DevOps

Combination of:

- Development
- Operations

👉 Faster software delivery

6. IoT in Cloud

Connecting smart devices to cloud for data processing.

7. Cloud AI

Cloud used for training AI/ML models.

8. Multi-cloud

Using multiple cloud providers together.

🎯 FINAL VIVA READY SUMMARY

If examiner asks:

“Explain Cloud Computing important topics”

Say:

👉 Cloud computing provides services like IaaS, PaaS, SaaS over the internet. It has deployment models like public, private, hybrid. Virtualization enables multiple VMs on one system. Security ensures data protection using encryption and access control. Advanced technologies include edge computing, serverless computing, and containerization.